

# Ranul Vithanage

Doctoral Student, Imperial College London, London, United Kingdom

r.thalahitiya-vithanage25@imperial.ac.uk — [linkedin.com/in/ranul-vithanage](https://www.linkedin.com/in/ranul-vithanage) — [ranulv.github.io](https://ranulv.github.io)

## RESEARCH INTERESTS

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Tutoring Robots, Skill Modelling, Human-Robot Interaction, Child-Robot Interaction, Human-Computer Interaction, Learning Science, Cognitive Science, Artificial Intelligence

## EDUCATION

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**Imperial College London**, London, UK

PhD in Computing

(Human-Robot Interaction)

Starting on 30.09.2025

Personal, Assistive, and Intelligent Robotics (PAIR Lab)

Supervised by Dr. Nicole Salomons

**University of Moratuwa**, Moratuwa, Sri Lanka

BSc Eng (Hons) in Mechanical Engineering

Dean's List in Semesters 1, 2, 7, and 8

2020 — 2024

Overall GPA: 3.77/4.20

First Class

## PUBLICATIONS

[ranulv.github.io/publications](https://ranulv.github.io/publications)

### Smart Music Therapist 1.0: Rhythmic Auditory Stimulation Integrated Robotic Walker as a Therapeutic Companion for Gait Rehabilitation

(Presented at IEEE Int. Conf. on Robot and Human Interactive Communication (RO-MAN) 2024) [*IEEE Xplore*] [*Paper*]

- Robot-facilitated Music Therapy towards a semi-autonomous solution for gait rehabilitation
- Telerehabilitation platform for Music Therapist to remotely monitor and consult patients
- Intelligent companionship (voice interaction, graphical user interface, music recommendation, etc.) integrated with a robotic walker fusing rehabilitation and socially assistive robotics

### Robust Autonomous Summoning Capabilities for a Patient Assistive Mobile Robot

(Presented at IEEE Int. Conference on Advanced Robotics and Mechatronics (ICARM) 2024) [*IEEE Xplore*] [*Paper*]

- Robust navigation without reliance on Indoor Positioning Systems (IPS) for infrastructure-limited areas
- LiDAR-based SLAM for Localization and Mapping
- Sound Source Localization for out-of-line-of-sight navigation and Gesture Recognition with Depth estimation by Stereo Vision for line-of-sight navigation

### Design and Simulation of a Thermally Actuated Microgripper with a Compliant Bistable Release Mechanism for Biomanipulation

10.1109/MERCon63886.2024.10689058

(Presented at Moratuwa Engineering Research Conference (MERCon) 2024)

[*IEEE Xplore*] [*Paper*]

- MEMS-based Microgripper with an Active Plunger working on the principle of Bistable Compliant Mechanism
- Featuring Double and Cascaded V-shaped (Chevron-shaped) actuators for effective manipulation
- Simulation results of displacement and temperature in the range preferred for bio samples

### Intelligent Controller for Active Walking Guidance of a Robotic Walker

(Submitted to IEEE/RAS-EMBS International Conference on Rehabilitation Robotics 2025)

[*Pre-print*]

- Active Navigation Control Interface using armrest-mounted force-sensitive resistor matrix
- Comparative evaluation of Rule-based and Fuzzy Control Methods on intuitiveness and smoothness of control
- Armrest-integrated Haptic feedback for direction-specific obstacle avoidance

## TEACHING EXPERIENCE

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### Graduate Teaching Assistant

Oct 2025 — Present

*Department of Computing, Imperial College London, UK*

- Conducting lab sessions of the modules COMPX0120: Fundamentals and Innovative Applications of Robotics, COMP 60006/70089: Computer Vision, COMP 60019/70084: Robotics
- Evaluating the module coursework of COMP 40012/40018: Discrete Mathematics, Logic & Reasoning

### Temporary Lecturer

Jan 2025 — Jul 2025

*Department of Mechanical Engineering, University of Moratuwa, Sri Lanka*

- Conducting lectures in the modules of Automotive Engineering and Design of Machine Elements
- Conducting Continuous Assessments including tutorials, practicals and Final Paper Evaluations
- Evaluating undergraduate viva, presentations, assignments, and reports.

### Temporary Instructor

Jul 2024 — Jan 2025

*Department of Mechanical Engineering, University of Moratuwa, Sri Lanka*

- Conducting practical lab sessions in the modules of Mechanics, Mechanics of Materials, and Manufacturing Processes for Mechanical Engineering undergraduate students
- Conducting Familiarization sessions for MATLAB, COMSOL, SOLIDWORKS software
- Evaluating undergraduate viva, presentations, assignments, and reports.

### Visiting Instructor

Apr 2024 — Jun 2024

*Department of Mechanical Engineering, University of Moratuwa, Sri Lanka*

- Conducted practical lab sessions under the module Fundamentals of Mechatronics
- Facilitated workshops on Single-Board Computers (Raspberry Pi), Micro-controllers (Arduino), Image Processing principles and tools for Robotics projects

## AWARDS

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### Student Travel Grant (USD 2000)

Aug 2024

*IEEE Int. Conference on Robot and Human Interactive Communication (RO-MAN) 2024*

Pasadena, California

## VOLUNTEERING EXPERIENCE

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### Diriya Arunalu [[Link to Facebook Page](#)]

*Co-Founder & Educator*

Galle, Sri Lanka

Jan 2019 - Dec 2019

- Co-founded the initiative to support Mathematics education in underprivileged schools in Galle, Sri Lanka
- Organized numerous teaching sessions in correspondence with school management
- Taught Mathematics to students in Grades 10 and 11 preparing for the G.C.E. O/L Examination
- Organized book donations among peers to donate to ill-resourced libraries in the local area

## REFERENCES

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### Dr. Nicole Salomons

*Assistant Professor, Department of Computing, Imperial College London, UK*

E-mail: [n.salomons@imperial.ac.uk](mailto:n.salomons@imperial.ac.uk)

### Dr. Daniel Tozadore

*Lecturer, Department of Computer Science, University College London, UK*

E-mail: [d.tozadore@ucl.ac.uk](mailto:d.tozadore@ucl.ac.uk)